Can Large Language Models Identify Materials from Radar Signals?

Jiangyou Zhu
The Chinese University of Hong Kong
Hong Kong SAR, China
zj124@ie.cuhk.edu.hk

Hongyu Deng
The Chinese University of Hong Kong
Hong Kong SAR, China
dh021@ie.cuhk.edu.hk

He Chen*
The Chinese University of Hong Kong
Hong Kong SAR, China
he.chen@ie.cuhk.edu.hk

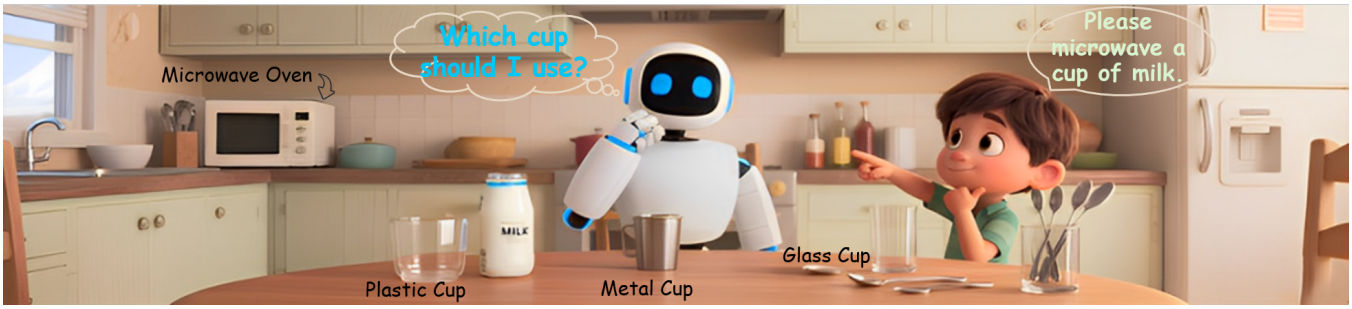


Figure 1: Home robots often need to distinguish the material composition of visually similar objects during everyday tasks. This figure illustrates such a scenario, where the robot needs to identify a glass cup to safely microwave a cup of milk, a task made challenging by its visual similarity to a plastic cup, which may not be microwave-safe. Consequently, additional sensing modalities or methods are necessary to achieve accurate material identification. This image is generated by the Doubao AI model from a sketch input, with additional text overlaid on the generated image.

Abstract

Accurately identifying the material composition of objects is a critical capability for AI robots powered by large language models (LLMs) to perform context-aware manipulation. Radar technologies (e.g., millimeter-wave radar) offer a promising sensing modality for material recognition tasks, providing robustness to lighting conditions and high spatial resolution. When combined with deep learning, radar technologies have demonstrated strong potential in identifying the material composition of various objects. However, existing radar-based solutions are often constrained to closed-set object categories and typically require task-specific data collection to train deep learning models, largely limiting their practical applicability. This raises an important question: *Can we leverage the powerful reasoning capabilities of pre-trained LLMs to directly infer material composition from raw radar signals*? Answering this question is non-trivial due to the inherent redundancy of radar signals and the fact that pre-trained LLMs have no prior exposure to raw radar data during training. To address this, we introduce LLMaterial, the first study to investigate the feasibility of using LLMs to identify materials directly from radar signals. First, we introduce a physics-informed signal processing pipeline that distills high-redundancy radar raw data into a set of compact intermediate parameters that encapsulate the material's intrinsic characteristics. Second, we adopt a retrieval-augmented generation (RAG) strategy to provide the LLM with domain-specific knowledge, enabling it to interpret and reason over the extracted intermediate parameters. Leveraging this integration, the LLM is empowered to perform step-by-step reasoning on the condensed radar features, achieving open-set material recognition directly from raw radar signals. Preliminary results show that LLMaterial can effectively distinguish among a variety of common materials, highlighting its strong potential for real-world material identification applications.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**.

Keywords

Material identification, radar sensing, large language model, retrieval-augmented generation.

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1 Introduction

Object recognition is a fundamental capability for autonomous systems, underlying a wide range of applications in home robotics and industrial automation [13]. In recent years, the application of AI to vision-based object recognition has made substantial strides.

*He Chen is the corresponding author.

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In particular, the rise of large language models (LLMs) has led to remarkable results not only in object identification but also in recognizing object attributes such as material composition [2, 11, 12]. This progress is largely attributed to a key strength of LLMs: their deep semantic understanding, developed through large-scale pretraining on diverse and extensive datasets. However, our real-world experiments reveal a critical limitation: LLMs relying solely on visual inputs often fail to distinguish between visually similar objects, which can lead to unsafe decisions. For instance, a robot may be unable to differentiate between a glass and a plastic cup, an error that poses a safety risk when heating liquids, as only glass is microwave-safe (see Figure 1). This challenge has also been highlighted in prior work [1], emphasizing the need for more robust recognition systems that extend beyond vision alone.

As an emerging sensing modality, radar is increasingly recognized for its noticeable potential in material recognition. Specifically, RadarCat [10] pioneered this direction by using time-domain analysis of millimeter-wave (mmWave) signals and machine learning to classify materials. mSense [7] quantitatively characterizes the material's reflectivity using the material reflection feature (MRF), and then determines the material type by finding the best match against a pre-trained database. CRFUSION [8] combines camera and radar to simultaneously identify object categories and materials, using the energy reflection factor (ERF) as a key feature within a modality fusion network. However, a common limitation of these approaches is that they often rely on extensive data collection to train end-to-end material identification models. However, their effectiveness is typically confined to a narrow range of materials, which noticeably limits their practical utility in real-world applications. Given the profound comprehension capabilities of LLMs, a promising direction is to leverage them for interpreting radar signals in material recognition tasks. This leads us to our research question: *Can LLMs infer an object's electromagnetic properties and accurately identify its material from radar signals in an open-set setting?*

Answering this question is nontrivial, as it involves overcoming challenges stemming from both the limitations of the LLM's prior knowledge and the complex nature of raw radar data. First, because radar raw data is rarely represented in the pretraining datasets of LLMs, these models lack the inherent ability to directly interpret the semantic content embedded in radar signals. This necessitates a dedicated preprocessing step to translate raw radar inputs into a representation that aligns with the LLM's modality and semantic understanding. In addition, the high dimensionality and inherent redundancy of raw radar data present significant challenges. Much of the signal may be unrelated to the material's intrinsic properties, introducing noise that can obscure relevant information. As a result, a key challenge lies in guiding the LLM to focus selectively on material-relevant features within the radar signal.

This paper introduces LLMaterial, the first system that can leverage pre-trained LLMs to directly identify materials from raw radar signals. Our contributions in developing LLMaterial are threefold:

- We develop a physics-informed signal processing pipeline to extract a material's intrinsic dielectric constant from raw radar signals, enabling tractable and interpretable inputs for LLM.
- We incorporate a retrieval-augmented generation (RAG) framework to equip the LLM with domain-specific knowledge, enabling it to develop a chain-of-thought (CoT) to reason about properties.
- We perform an initial validation to assess the practical feasibility of LLMaterial. This validation underscores the promise of our approach and provides a strong foundation for future research toward more intelligent and context-aware assistive technologies.

We note that recent studies have explored the use of LLMs to interpret data from other non-visual sensors, such as inertial measurement units (IMUs) [4, 6] and ultra-wideband (UWB) sensors [9].

2 System Design

In this section, we introduce the architecture of LLMaterial, designed to perform robust material identification through a new integration of radar sensing and LLM-based reasoning. In our work, we used an mmWave radar sensor (IWR6843ISK) manufactured by Texas Instruments (Dallas, TX, USA). The overall workflow of LLMaterial is depicted in Figure 2. For visually similar objects, LLMaterial first acquires the raw target echo signal using an mmWave radar. Next, this raw radar data is preprocessed, followed by performing electromagnetic parameter estimation to extract key parameters. These parameters are then fed into an LLM-based system augmented with specialized domain knowledge. Leveraging RAG, the model then generates a CoT process to reason about all electromagnetic properties, ultimately leading to robust material identification.

2.1 Radar Signal Processing

This part details our signal preprocessing pipeline. We first perform multichannel accumulation on the 3D radar echoes to generate a high-resolution range-Doppler (RD) map. Subsequently, the range-angle (RA) map is derived by processing the data in the slow-time dimension. The combination of these maps provides the target's distance (R), velocity (V), and angle (θ) relative to the radar, which are crucial for its accurate localization. Representative RD and RA maps are depicted in Figure 3.

To enhance the precision of target angle estimation, we employ beamforming techniques to accurately resolve the target's direction of arrival (DoA). This step is crucial, as the antenna array's native resolution is limited by its physical aperture; a smaller aperture results in a wider half-power beamwidth (HPBW). As shown in Figure 4, the HPBW of our mmWave radar array was measured to be $12.8°$, and the target angle after beamforming can achieve better accuracy.

The aforementioned steps constitute our signal preprocessing stage. Once a target is detected and its signal is extracted, the subsequent analysis focuses on electromagnetic parameter estimation, as shown in Figure 5. First, for a target, the received echo power is determined by the classical radar equation, expressed as:

$$P_r = P_t G_t G_r \frac{\lambda^2 \sigma}{(4\pi)^3 R^4}, \tag{1}$$

where P_r is the received power, P_t is the transmitted power, G_t and G_r are the gains of the transmitting and receiving antennas, respectively, λ is the signal wavelength, σ is the target's RCS, and R is the distance of the target relative to the mmWave radar. Based on the expression of P_r, the signal-to-noise ratio (SNR) of the target echo signal can be expressed as:

$$SNR = \frac{P_r}{P_n} = \frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 k T_n B R^4} = K \frac{\sigma}{R^4}, \tag{2}$$

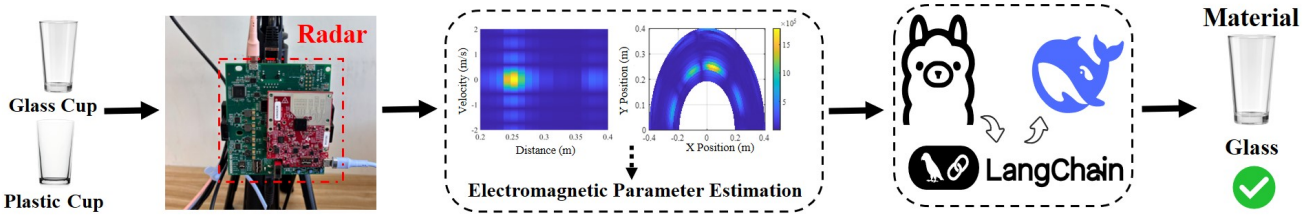


Figure 2: Overview of LLMaterial.

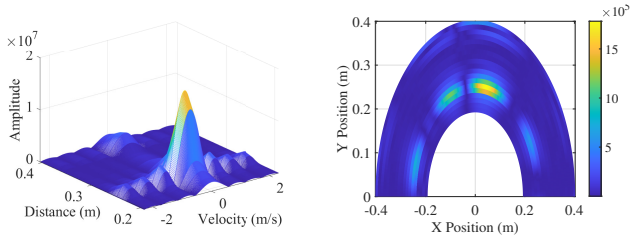


Figure 3: (Left) RD map; (Right) RA map.

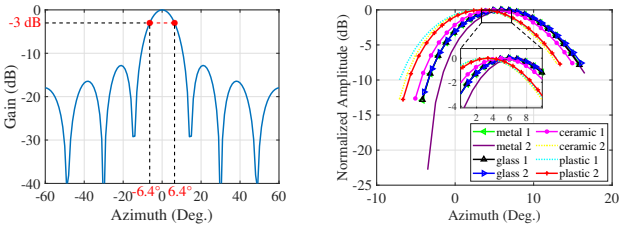


Figure 4: (Left) Beam Pattern of the Antenna Array; (Right) Beam Direction After Beamforming.

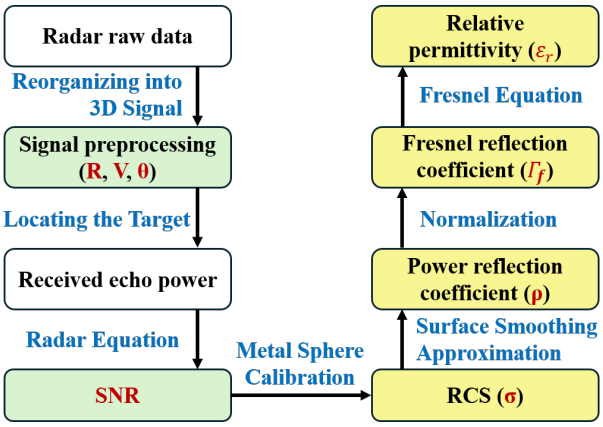


Figure 5: Workflow of radar signal processing.

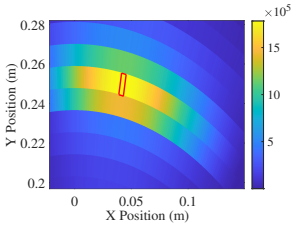


Figure 6: PRCA (red box).

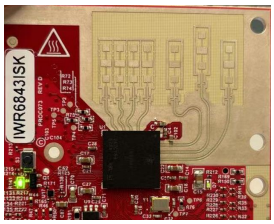


Figure 7: Antenna array.

where P_n is the total noise power of the receiver. For thermal noise, $P_n = kT_nB$, where k is the Boltzmann constant, T_n is the effective system noise temperature, and B is the bandwidth. The constant K combines all system-specific parameters, highlighting that the SNR is primarily dependent on the target's RCS and its distance. Constant K can be expressed as:

$$K = \frac{P_t G_t G_r \lambda^2}{(4\pi)^3 k T_n B}. \tag{3}$$

To determine the system constant K, we performed a calibration using a metal sphere ($d = 63$ mm) as a standard target. For a sphere with a diameter significantly exceeding the signal wavelength ($\lambda = 5$ mm), the sphere's RCS in the optical scattering regime equals its physical cross-sectional area, which can be calculated as:

$$\sigma_c = \pi \left(\frac{d}{2}\right)^2 \approx 0.0031\,\mathrm{m}^2. \tag{4}$$

After measuring the SNR of the sphere, which serves as a known reference, we computed K by inversion of Equation (2). This calibrated constant enables the direct calculation of an unknown target's RCS, which is given by:

$$\sigma = SNR\frac{R^4}{K}. \tag{5}$$

However, the RCS itself is a composite parameter, integrating the intrinsic electromagnetic properties of the target's material with its overall geometric shape. To disentangle these attributes and analyze the material characteristics more directly, we relate the RCS to the backscattering coefficient. In the field of radar, the backscattering coefficient, denoted as σ_0, is defined as the average RCS per unit area, where A_c denotes the area on the ground associated with that object. The RCS of a specific object can be expressed as:

$$\sigma = \sigma_0 A_c. \tag{6}$$

Inspired by this principle, we shift our focus from the total RCS to the dominant reflection center that generates the signal peak. We define a power reflection coefficient ρ to represent the strong reflection signal originating from a small but highly reflective region, which we term as peak reflection cell area (PRCA) A_r. The extracted PRCA is shown in Figure 6. By approximating PRCA as a local specular reflector, ρ becomes analogous to σ_0. Therefore, RCS can be further expressed as:

$$\sigma = \rho A_r. \tag{7}$$

Theoretically, the power reflection coefficient ρ is described by the Fresnel equations, which define it as a function of the relative permittivity (ε_r) of the material, the polarization of the wave, and the angle of incidence. For the positive ρ that we calculate, we perform a normalization to obtain the Fresnel reflection coefficient Γ_f for vertical polarization. This step is justified by our core assumptions: specular reflection from a locally smooth surface combined with the known vertical polarization of our mmWave radar, as shown in Figure 7 (as solving for ε_r with horizontal polarization is mathematically intractable, resulting in a high-order polynomial, and our approach is consistent with commercial radars, which are commonly vertically polarized, thereby focusing our analysis to the vertical polarization case). Consequently, Γ_f is modeled as:

$$\Gamma_f = \frac{\varepsilon_r \cos\theta - \sqrt{\varepsilon_r - \sin^2\theta}}{\varepsilon_r \cos\theta + \sqrt{\varepsilon_r - \sin^2\theta}}, \tag{8}$$

where $\varepsilon_r = \varepsilon' - j\varepsilon''$, ε' is the real part, and ε'' is the imaginary part.

We define the variables as follows:

$$A = \varepsilon_r \cos\theta, \quad B = \sqrt{\varepsilon_r - \sin^2\theta}. \tag{9}$$

Then, we get:

$$\Gamma_f = \frac{A - B}{A + B}. \tag{10}$$

We can further rewrite it as:

$$\Gamma_f(A + B) = A - B. \tag{11}$$

Next, we expand and rearrange to obtain:

$$\begin{aligned} \Gamma_f A + \Gamma_f B &= A - B, \\ \Gamma_f A - A &= -\Gamma_f B - B, \\ A(\Gamma_f - 1) &= -B(\Gamma_f + 1). \end{aligned} \tag{12}$$

Substituting A and B back into Equation (12), we have:

$$\varepsilon_r \cos\theta(\Gamma_f - 1) = -\sqrt{\varepsilon_r - \sin^2\theta}(\Gamma_f + 1). \tag{13}$$

To proceed, we square both sides, expand and rearrange, and move all terms to the left side, getting:

$$\varepsilon_r^2 \cos^2\theta(\Gamma_f^2 - 2\Gamma_f + 1) - \varepsilon_r(\Gamma_f + 1)^2 + \sin^2\theta(\Gamma_f + 1)^2 = 0. \tag{14}$$

We treat Equation (14) as a quadratic equation in terms of ε_r, and then we can solve it using the quadratic formula. Due to space constraints, we do not provide an extensive derivation here. The final relationship between ε_r, Γ_f and θ is given by:

$$\varepsilon_r = \frac{(\Gamma_f + 1)^2 \left[1 \pm \sqrt{1 - \left(\frac{\sin 2\theta(\Gamma_f - 1)}{\Gamma_f + 1}\right)^2}\right]}{2\cos^2\theta[\Gamma_f - 1]^2}. \tag{15}$$

According to the principles of physics, for ε_r, which is a complex number, we choose the positive root. Specifically, when $\theta = 0$ (i.e., the target is located directly in front of the radar), we can get:

$$\varepsilon_r = \left(\frac{1 + \Gamma_f}{1 - \Gamma_f}\right)^2. \tag{16}$$

In summary, we have systematically traced the signal processing workflow from initial raw signal to SNR, then through the analysis of RCS (σ), power reflection coefficient (ρ), and Fresnel reflection coefficient (Γ_f), ultimately leading to the extraction of the target's fundamental material property: the relative permittivity (ε_r). This signal processing pipeline facilitates the extraction of essential physical characteristics of an object directly from raw radar signals.

2.2 RAG-Enhanced LLM

In this section, we present our LLM-based framework for material inference. As illustrated in Figure 8, the LLM interprets eight parameters extracted from the radar signal processing pipeline to predict the material composition of the target object. These include the target's basic information (R, V, θ) and SNR, along with four key electromagnetic features: σ, ρ, Γ_f, and ε_r.

As shown in Figure 8, we apply the RAG technique to enhance the LLM's material identification capabilities. This process begins by segmenting documents into chunks, which are embedded and indexed in a vector database. Once ready, LLMaterial performs a semantic search to retrieve the top-k relevant chunks. These, along with engineered prompts, inform the LLM, enabling it to generate accurate, traceable responses. This ensures that LLMaterial provides correct, trustworthy, and verifiable recognition results.

As shown in Figure 9, the online identification system of our proposed LLMaterial is divided into two main panels:

- **Left Panel (Radar Measurement Inputs):** This panel displays eight key parameters from the mmWave radar signal processing workflow, serving as quantitative inputs for subsequent analysis.
- **Right Panel (Online analysis of LLM):** This panel provides a guided two-step analytical process.
  (1) **Step 1: Upload Knowledge Base.** Users upload domain-specific documents to build the RAG system's knowledge base. The interface then visualizes the workflow, ensuring transparency and easy monitoring of the retrieval and augmentation process.
  (2) **Step 2: Start Analysis.** After loading the knowledge base, users can initiate analysis. The system processes the eight parameters, enabling the LLM to use RAG and CoT for comprehensive analysis and generating the final material identification result.

Moreover, Figure 9 shows the noticeable impact of RAG's knowledge base on the LLM's reasoning. As an initial effort to provide domain-specific context, our knowledge base was constructed from primary sources on key topics, such as RCS [3] and the relationship between dielectric constants and material properties [5].

3 Preliminary Experimental results and analysis

The parameters of mmWave radar are listed in Table 1. Our signal processing pipeline was implemented in MATLAB R2023a. The 14-billion-parameter DeepSeek LLM (deepseek-r1:14b) was employed.

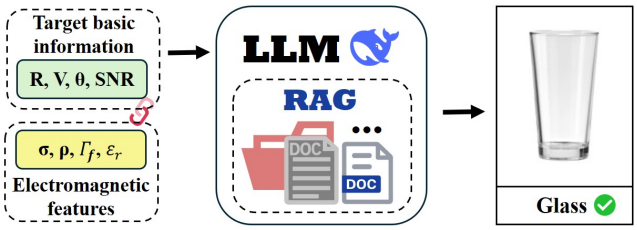


Figure 8: Workflow of the RAG-enhanced LLM for Radar-based material identification system.

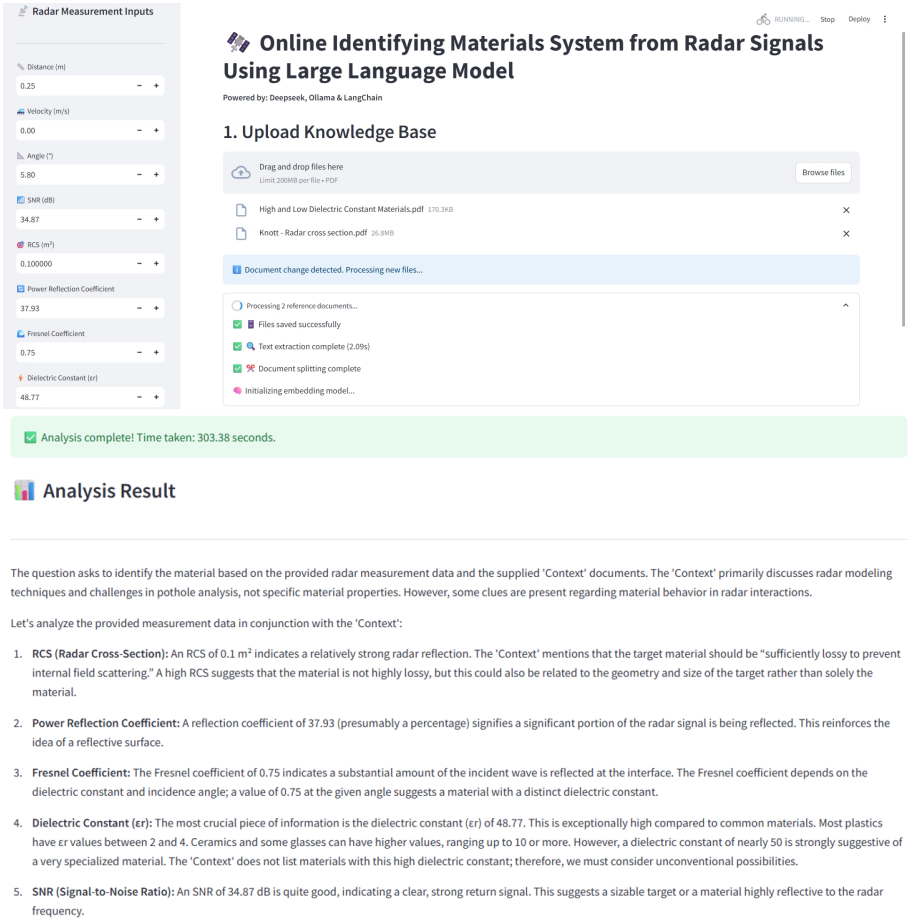


Figure 9: Online material identification from radar signals using our LLM-based system. (Up) Step 1; (Down) Step 2.

Table 1: Radar Parameter Settings.

Parameter	Value	Explanation
f_0	60 GHz	Carrier Frequency
S	66 MHz/μs	Modulation Slope
B	3.96 GHz	Bandwidth
f_s	10 MHz	Sampling Rate

All experiments were conducted on a Windows desktop equipped with a 2.60 GHz Intel Core i7-9750H CPU and 16 GB of RAM.

Experimental results: To evaluate LLMaterial's performance, we tested seven common objects made from four distinct material categories, with results summarized in Figure 10. The preliminary findings show that the electromagnetic parameters have unique signatures for each category, allowing the RAG-enhanced LLM to conduct a step-by-step CoT reasoning process. The final recognition results are accurate and consistent with the actual materials, validating LLMaterial and confirming that it is indeed feasible to apply LLMs to identify materials from raw radar signals.

Analysis: To validate the effectiveness of our proposed LLMaterial, we performed an ablation study. As detailed in Table 2, we

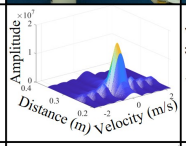
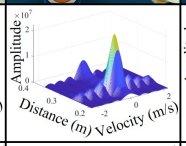
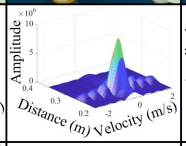
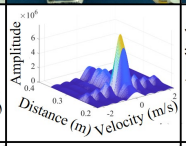
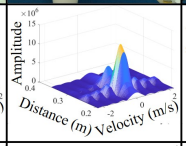
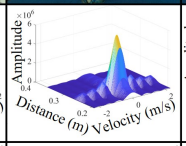
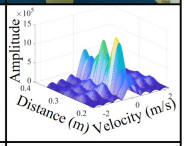
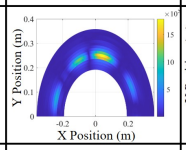
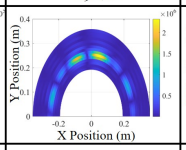
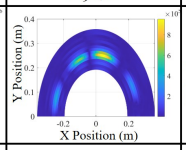
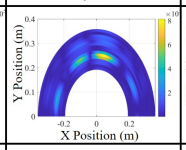
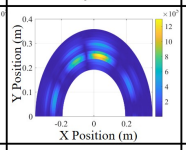
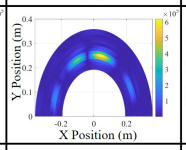
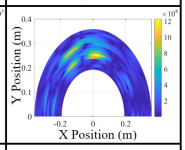
Scenario							
RD Map							
RA Map							
R (m)	0.2477	0.2477	0.2477	0.2477	0.2477	0.2477	0.2477
V (m/s)	0.00453	0.00453	0.00453	0.00453	0.00453	0.00453	0.00453
θ (°)	5.8	6.5	5.8	6.1	4.9	2.9	2.9
SNR (dB)	34.87	37.27	29.57	28.28	37.95	30.06	13.08
RCS (m²)	0.000419	0.000561	0.000224	0.000193	0.000297	0.000236	0.000027
ρ	37.93	50.75	20.32	17.44	26.89	21.43	2.47
Γ_f	0.75	1	0.4	0.34	0.53	0.42	0.05
ε_r	48.77	Inf	5.51	4.23	10.75	6.14	1.22
Material	Metal	Metal	Glass	Glass	Ceramic	Ceramic	Plastic

Figure 10: Summary of radar signal processing and recognition results: 7 common objects and 4 material types.

Table 2: Ablation study.

Method	Metal bottle	Metal box	Glass bottle	Glass cup	Ceramic cup	Ceramic mug	Plastic bottle
LLMaterial w/o RAG	×	√	×	×	×	×	√
LLMaterial w RAG	√	√	√	√	√	√	√

compared our full system (LLMaterial w/ RAG) against an ablated version without this module (LLMaterial w/o RAG). The results reveal a dramatic performance gap. The ablated version (LLMaterial w/o RAG) struggled in correctly identifying only two of the seven objects. In contrast, the LLMaterial system, empowered by RAG, achieved a perfect score, correctly identifying all seven objects. This demonstrates that the RAG module is not only beneficial but also indispensable, as it supplies the critical context necessary for the LLM to accurately interpret radar signals, ultimately facilitating the precise identification of object materials and enhancing the overall effectiveness of the material identification process.

4 Conclusion and future work

In conclusion, this paper introduces LLMaterial, the first framework to successfully leverage LLMs for radar-based material identification. We tackle this task through two key innovations: (1) a physics-informed signal processing pipeline that transforms raw radar data into meaningful electromagnetic parameters, and (2) a RAG-enhanced LLM that interprets these parameters to infer the object's material composition. Our preliminary experiments confirm that LLMaterial achieves accurate material recognition, demonstrating for the first time that LLMs can effectively interpret object materials from raw radar signals, especially addressing the differentiation of materials in visually similar objects.

We acknowledge several limitations in the current study. This work serves as a preliminary validation, focusing on only four material types. Future research should pursue three key directions: (1) broadening the diversity of evaluated materials to verify generalizability; (2) enhancing the RAG knowledge base and incorporating advanced CoT reasoning strategies to boost interpretability and accuracy; and (3) developing robust radar-vision fusion techniques for multimodal material recognition in real-world scenarios.

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